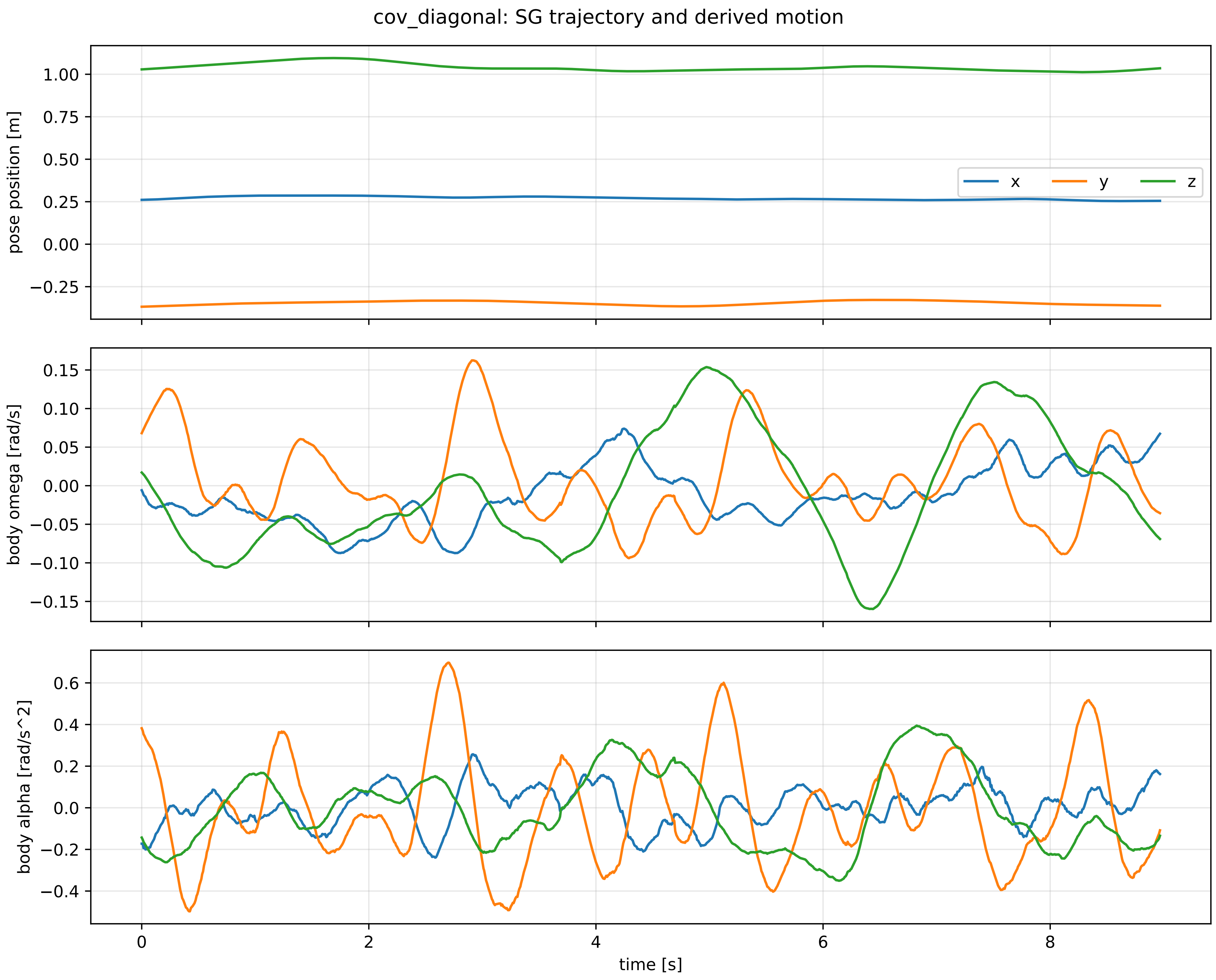
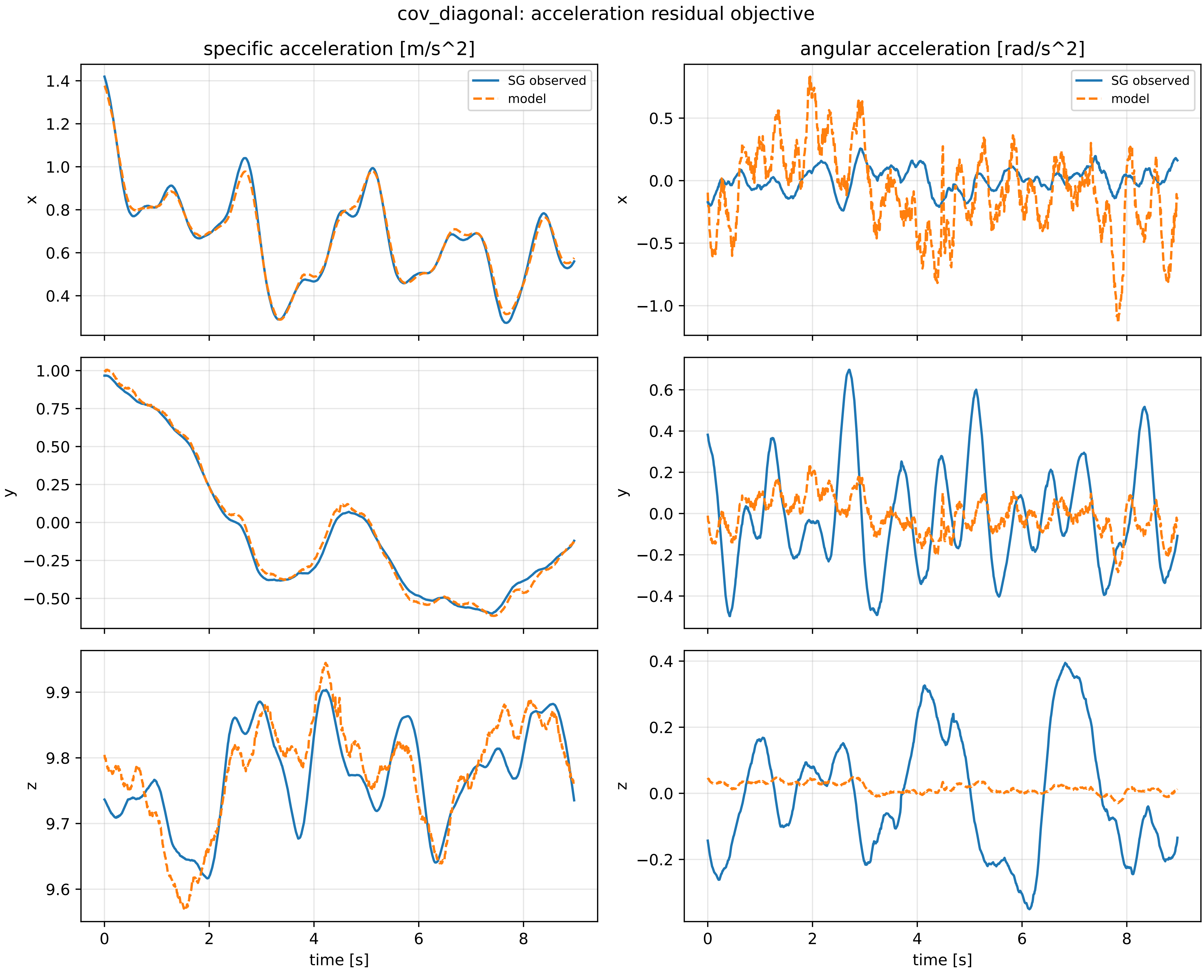
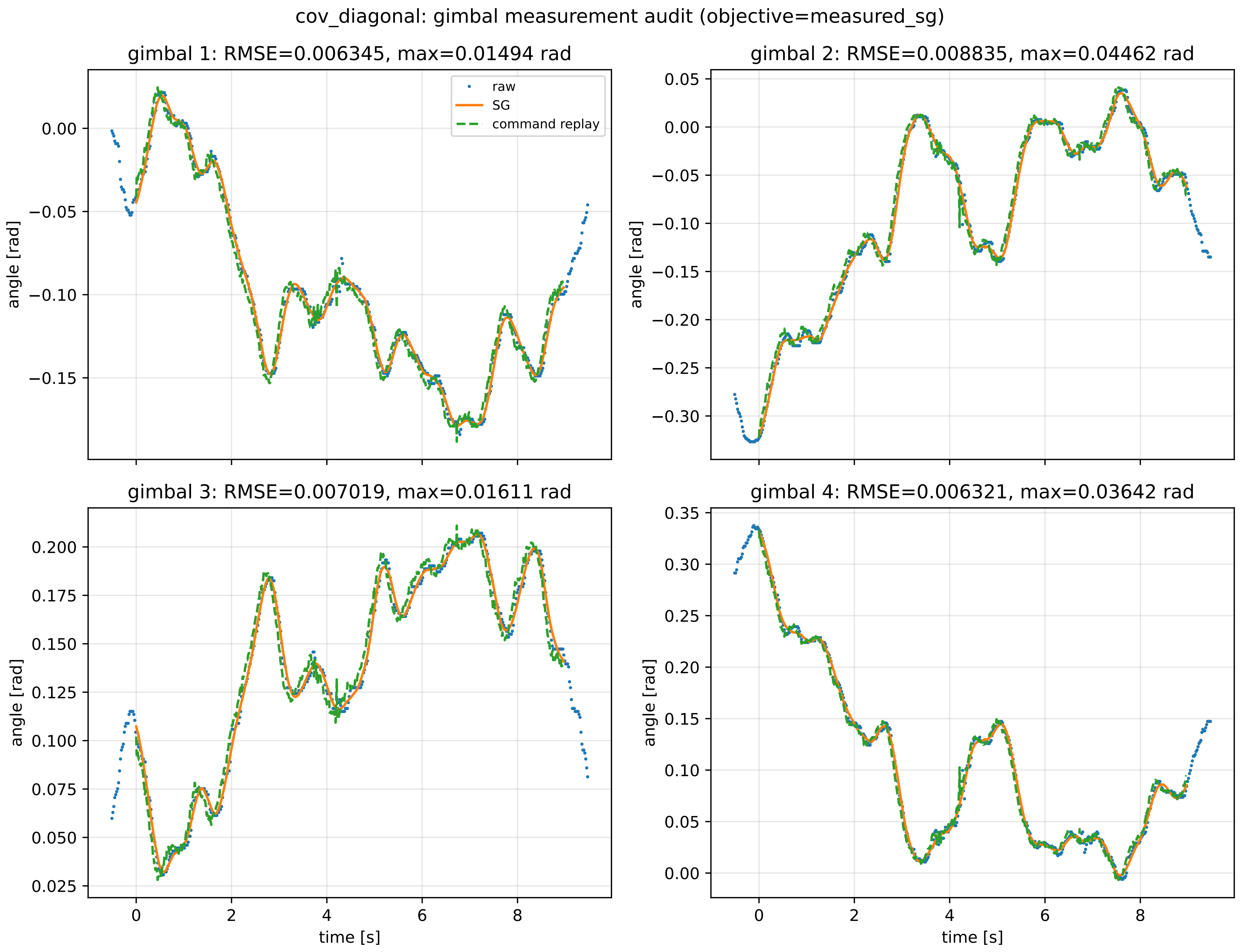
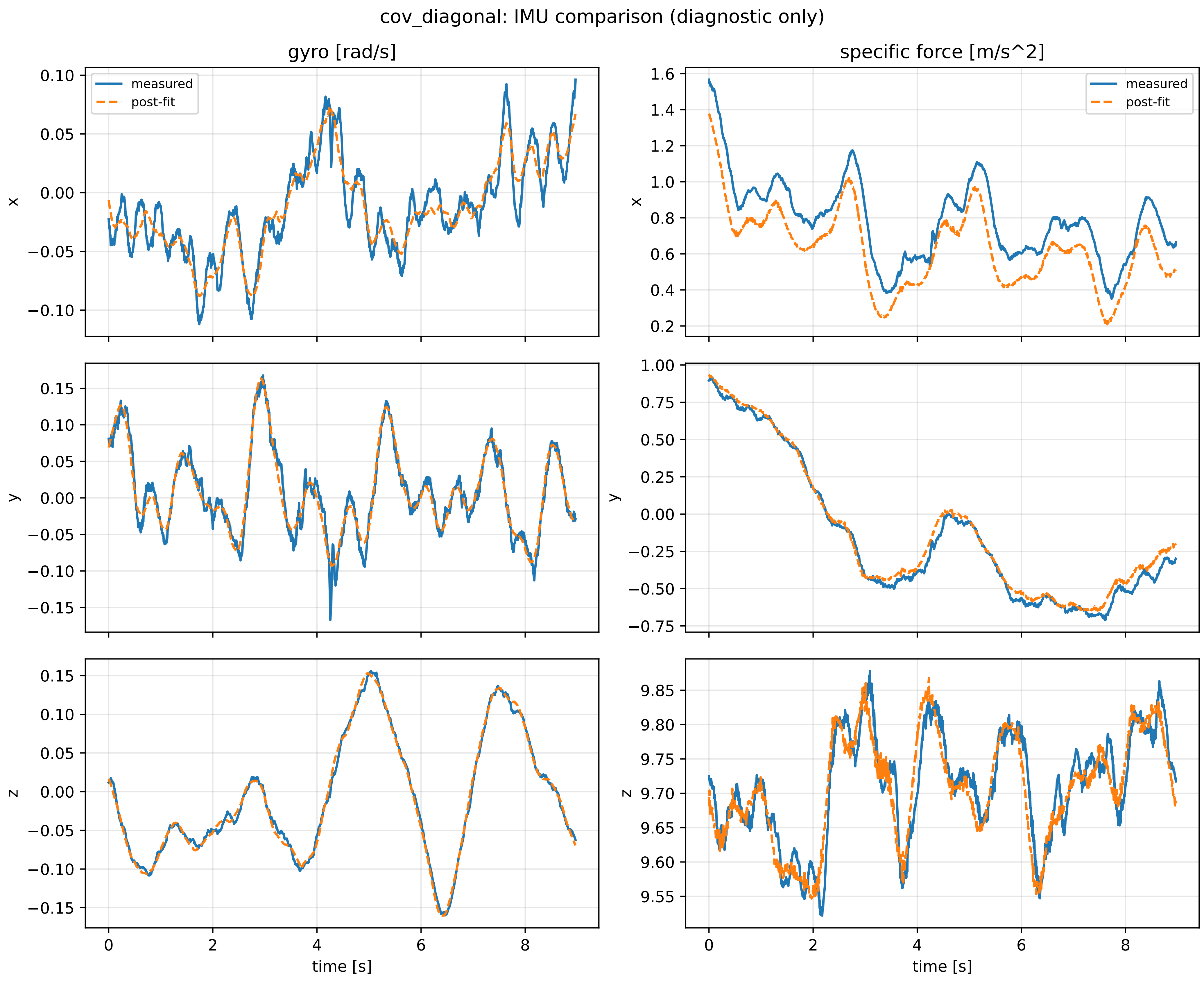


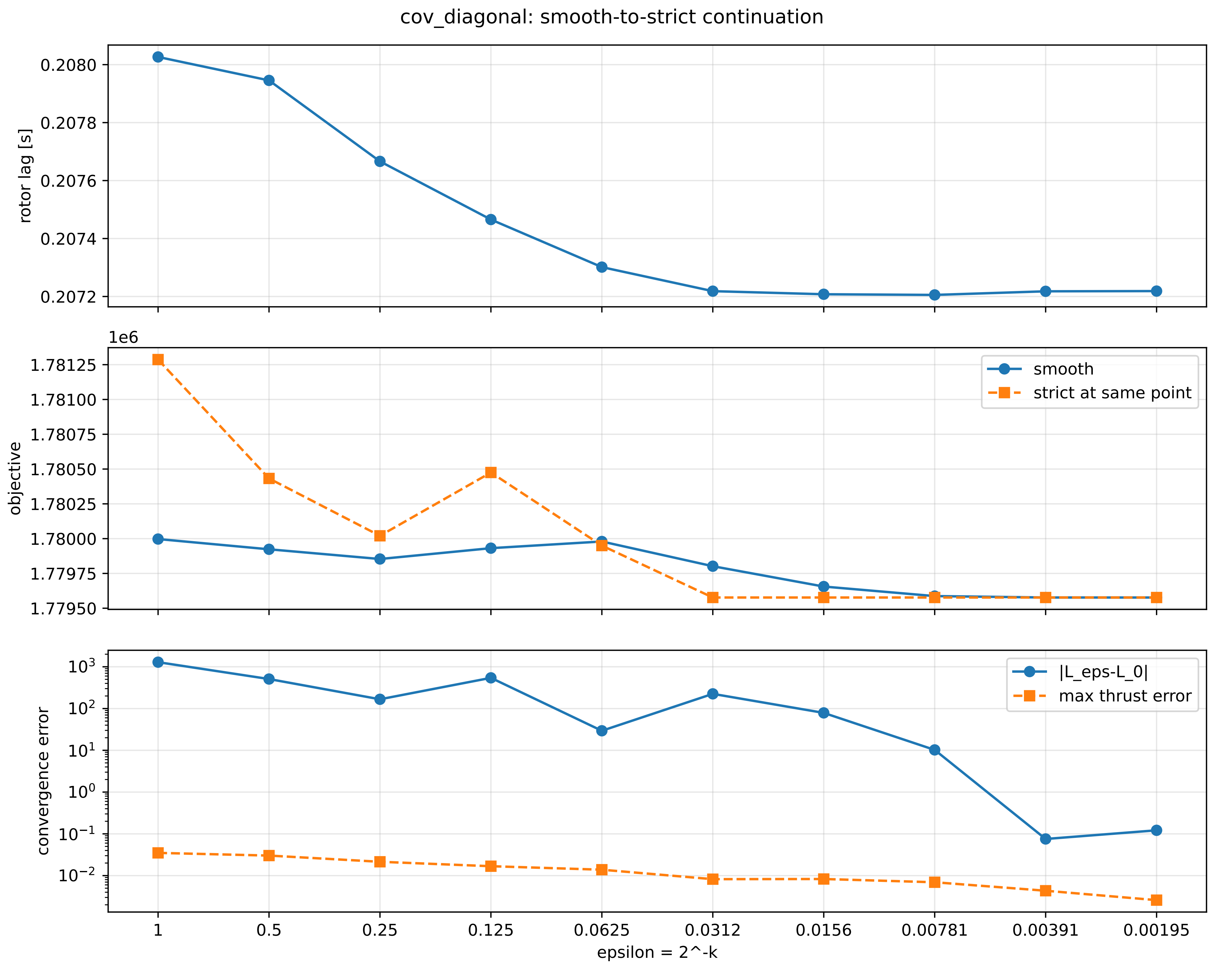
cov_diagonal: SG trajectory and derived motion
pose position [m]
1.00
0.75
0.50
0.25
0.00
−0.25
x
y
z
body omega [rad/s]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
body alpha [rad/s^2]
0.6
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8
time [s]

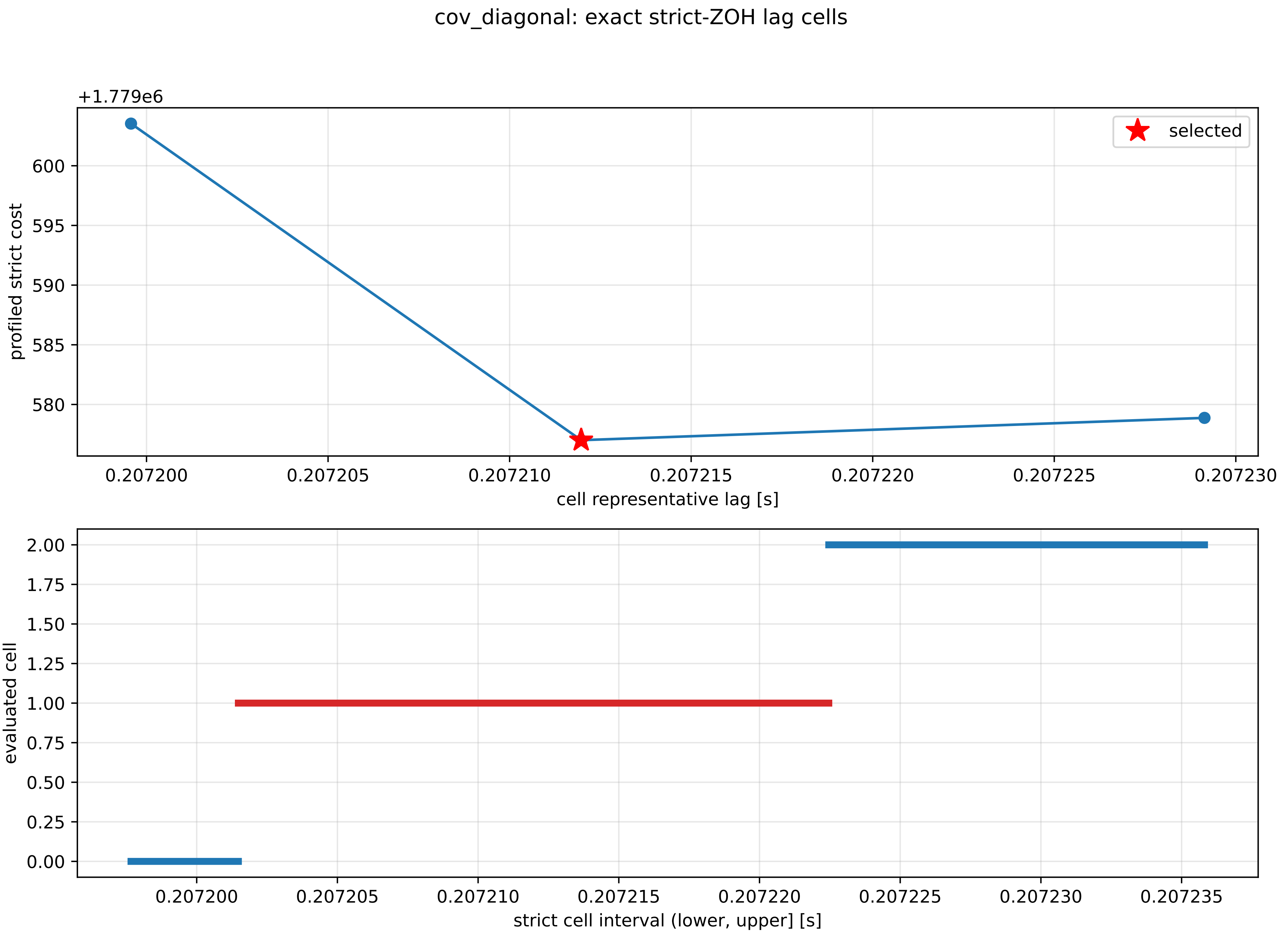

cov_diagonal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]


cov_diagonal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.006345, max=0.01494 rad
gimbal 2: RMSE=0.008835, max=0.04462 rad
gimbal 3: RMSE=0.007019, max=0.01611 rad
gimbal 4: RMSE=0.006321, max=0.03642 rad
raw
SG
command replay
angle [rad]
time [s]



cov_diagonal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]

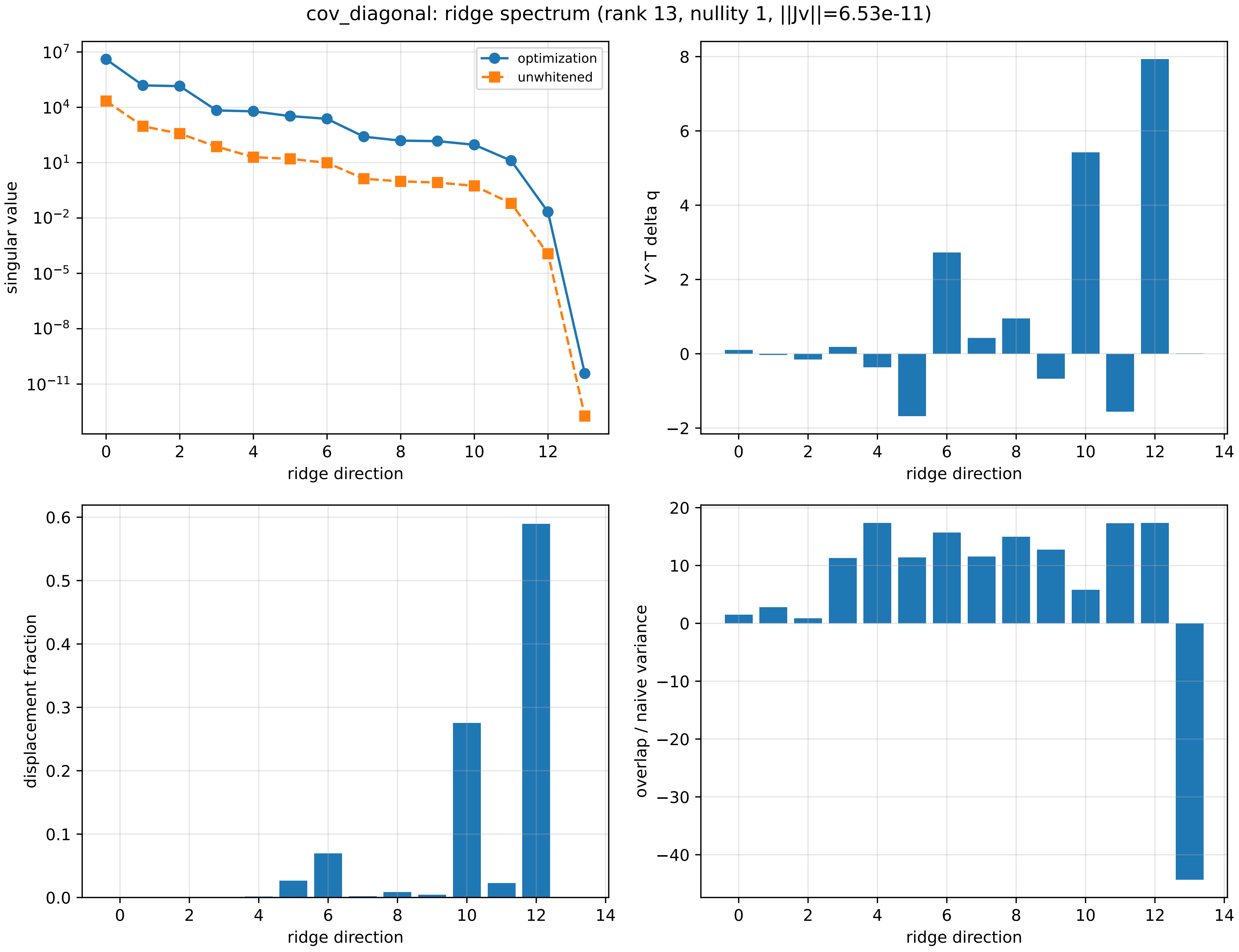

cov_diagonal: smooth-to-strict continuation
rotor lag [s]
0.2080
0.2078
0.2076
0.2074
0.2072
1e6
objective
1.78125
1.78100
1.78075
1.78050
1.78025
1.78000
1.77975
1.77950
smooth
strict at same point
convergence error
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k



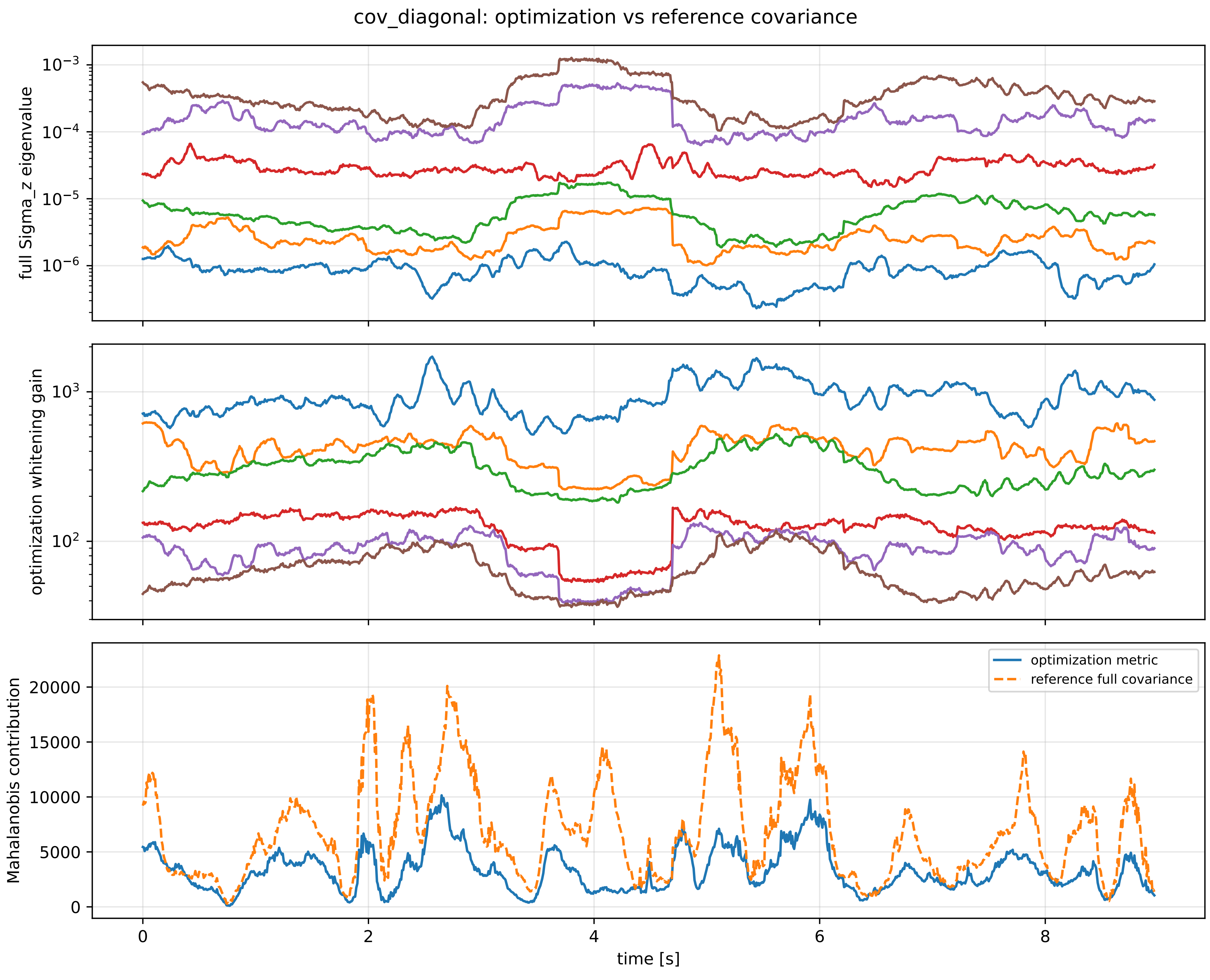
selected (0.207201481, 0.207222462] s; final smooth 0.207218685 s; data boundary=False

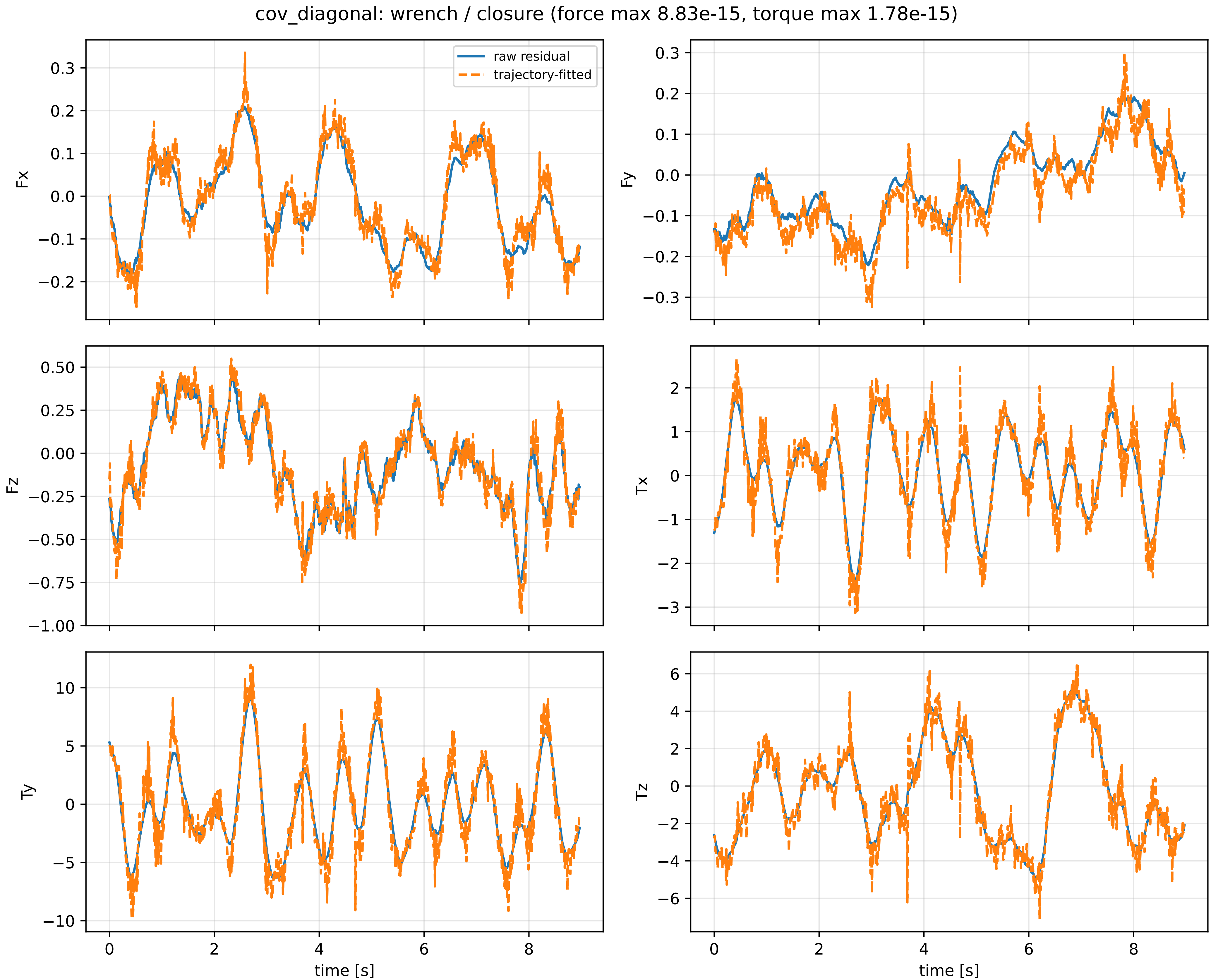
cov_diagonal: parameter estimate

```
Case: cov_diagonal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.13323107
inertia [kg m^2]:
[[ 0.9287348  -3.33260529 -0.35654212]
 [-3.33260529 12.79928616 -0.11728849]
 [-0.35654212 -0.11728849 13.66576885]]
CoG body [m]: [ 0.05644442 -0.16530322  0.02680258]
force effectiveness: [1.57197591 2.33949395 0.4824889  0.07581172]

scale-free J/m [m^2]:
[[ 0.2964144  -1.06363215 -0.11379375]
 [-1.06363215  4.08501189 -0.03743372]
 [-0.11379375 -0.03743372  4.36155794]]
scale-free f/m: [0.50171081 0.74667138 0.15399085 0.02419602]

rotor lag cell: (0.207201481, 0.207222462] s
representative: 0.207211971 s
```



cov_diagonal: ridge spectrum (rank 13, nullity 1, ||Jv||=6.53e-11)
optimization
unwhitened
singular value
ridge direction
V^T delta q
displacement fraction
overlap / naive variance


cov_diagonal: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-3}
10^{-4}
10^{-5}
10^{-6}
optimization whitening gain
10^{3}
10^{2}
Mahalanobis contribution
20000
15000
10000
5000
0
optimization metric
reference full covariance
0
2
4
6
8
time [s]


cov_diagonal: wrench / closure (force max 8.83e-15, torque max 1.78e-15)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]